LESSON PLAN

PROPELLER THEORY (7)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	General precautions to be observed while working on propeller, ground run
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	General precautions to be observed while working on propeller, ground run
5. Specific Objectives	7.1.2.1 Trainee will be able to explain propellers and its associated maintenance practice

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of propeller ground run safety checks, instructor supervision of trainee observation, safety precautions regarding blade clearance and personnel positioning
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary components of a propeller assembly?
(b) Why is a ground run performed before engine start?
(c) What is the standard safety distance for personnel during a propeller ground run?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Pre-flight visual inspection of propeller blades • Verification of propeller hub and spinner integrity • Clearance checks for ground run area • Engine start sequence and RPM monitoring • Post-run inspection for blade damage or imbalance	L & Ds	PPT, IB	15 min
(b)	• Safety protocols for personnel during ground run • Proper use of propeller brake and chocks • Identification of abnormal vibration or noise • Emergency stop procedures for ground run • Documentation of ground run results	L & Ds	PPT, PDF	20 min
(c)	• Maintenance practices for propeller blade balance • Torque specifications for propeller bolts • Storage and handling precautions for propellers • Inspection criteria for cracked or damaged blades • Reporting procedures for propeller defects	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) List three critical safety precautions during a propeller ground run.
(b) Explain the procedure for identifying a damaged propeller blade during inspection.
(c) Describe the correct steps to take if abnormal vibration occurs during a ground run.

12. Home Assignment and Follow Up	Prepare a brief report outlining the key differences between pre-flight propeller inspection and post-ground run inspection, referencing the next lesson plan (LP 02) on detailed propeller maintenance.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min